• **GAUGE—TEMPERATURE GAUGE FLUCTUATION AND/OR "KNOCKING/THUMPING" NOISE FROM COOLING SYSTEM—4.0L OHV ENGINE** • **COOLING SYSTEM—PULSATING COOLANT FLOW—NOISE—"KNOCKING/THUMPING"—AUDIBLE IN PASSENGER COMPARTMENT—4.0L OHV ENGINE** • **NOISE—"KNOCKING/THUMPING"—COOLING SYSTEM—AUDIBLE IN PASSENGER COMPARTMENT—4.0L OHV ENGINE**	**Article No. 99-23-8**

FORD: 1995-2000 EXPLORER, RANGER

This TSB article is being republished in its entirety to include 2000 model year vehicles and to provide Service Part(s) and Labor Time information.

ISSUE

A temperature gauge fluctuation and/or a "knocking/thumping" noise may be audible in the passenger compartment on some vehicles. This noise is most pronounced with the engine at normal operating temperature. The noise is caused by steam formation in the cooling system. The noise is then transmitted through the heater hoses and heater core.

ACTION

Install a Coolant By-Pass Kit. This increases coolant flow which reduces steam formation. Refer to the following Diagnostic and Service Procedure for details.

DIAGNOSTIC AND SERVICE PROCEDURE

1. Check for excessive exhaust system restrictions. A restricted exhaust could elevate cooling system temperatures.
2. If dash-mounted cooling temperature gauge fluctuates, check that gauge and sending unit are functioning properly.
3. Check for restrictions or kinks in the radiator or heater hoses.
4. Pressure check cooling system for leaks.
5. Confirm proper Engine Coolant Temperature (ECT) sensor signal to EEC-V processor.
6. Check for correct 50/50 coolant mix. Verify that freezing point is -37°C (±-15°C) or -35°F (±5°F).
7. If concern still exists, install service Coolant By-Pass Kit and verify that concern has been resolved. Refer to the Application Chart in this article.

Article No. 99-23-8 Cont'd.

COOLANT BY-PASS KIT CONTENTS					
Application	**Coolant By-Pass Kit (-8522-)**	**T-Fitting**	**By-Pass Tube**	**Hose Assembly**	**Instruction Sheet**
1997-2000 Ranger with automatic or manual transmission and 1997-2000 Explorer with automatic transmission	F8PZ-AA	Yes	Yes	Yes	Yes
1995-2000 Explorer with manual transmission and 1995-1996 Explorer with automatic transmission	F8PZ-BA	Yes	Yes	Yes	Yes
1995-1996 Ranger with automatic or manual transmission	F8PZ-CA	Yes	Yes	Yes	Yes

PART NUMBER	PART NAME
F8PZ-8522-AA	Coolant By-Pass Kit
F8PZ-8522-BA	Coolant By-Pass Kit
F8PZ-8522-CA	Coolant By-Pass Kit

OTHER APPLICABLE ARTICLES: 97-18-9
SUPERSEDES: 99-7-2
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
992308A	Install Service Coolant By-Pass Kit	0.7 Hr
992308B	For Diagnostics, Use "A" Time Per The Warranty And Policy Manual	

DEALER CODING

BASIC PART NO.	CONDITION CODE
8522	42

OASIS CODES: 204000, 204200, 208000, 402000, 497000, 702000, 702100, 702300